

Four Artifacts in Ablation-Based Causal-Effect Measurement of Sparse Autoencoder Latents

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Abstract

1 Introduction

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Setup. We measure the causal effect of a single SAE latent by ablating its contribution to the residual stream and reading the change in the model’s next-token distribution. For latent f with decoder direction d_f and activation a_f at position t , we replace $h_t \leftarrow h_t - a_f d_f$ at the SAE’s hook point and record

$$\text{KL}_t = D_{\text{KL}}(p(\cdot \mid h) \parallel p(\cdot \mid h')) , \quad (1)$$

evaluated at position t and, for the propagation analysis, at $t+1$ and $t+3$. All experiments use `gemma-2-2b` and residual-stream Gemma Scope dictionaries, plus a matched pair of TopK dictionaries trained here (§5).

The protocol. Each item below is an artifact established in §4, stated as the correction it implies. A practitioner reporting ablation-based causal effects should:

1. **Drop special-token positions** before any pooled statistic. Including position 0 moves the explained variance of a released Gemma Scope dictionary from 0.863 to -3.5 .
2. **Report effect per unit perturbation norm**, not raw. Perturbation magnitude varies from 3.60 to 14.76 across activation-frequency deciles, and ignoring it reverses the sign of the frequency conclusion.
3. **Report the per-latent nearest-neighbour cosine distribution**, and adjust or match on it *in dictionaries that exhibit near-duplicate structure*. Whether yours does is one line of code.
4. **Report unembedding alignment** alongside the effect. This is a sensitivity rather than a confound (§4), so both the unadjusted and alignment-adjusted estimates belong in the table.
5. **Report the jackknife curve**, not the mean or the variance ratio. Removing one latent of 240 moves our variance ratio from $42.1\times$ to $7.2\times$.
6. **State the sampling scheme.** The size of the alignment sensitivity depends on it, and no sample of a few hundred latents reaches the extreme tail of the alignment distribution.

Estimator. Group comparisons use the Hodges–Lehmann estimator, the median of pairwise ratios, reported with a bootstrap interval. HL is the point estimate belonging to the Mann–Whitney test; pairing a Mann–Whitney p -value with a ratio-of-medians interval, as we did initially, produces an apparent contradiction that is an artifact of mixing frameworks.

Convention. Ratios of the form KL_{t+k}/KL_t are computed *feature-level*: a per-latent median first, then the ratio of medians across latents. The observation-level alternative differs by a non-constant factor 0.97–1.35 and is not used.

4 Artifacts

Each subsection gives the mechanism, the evidence, the control, and the effect on the reported number.

4.1 Special-token reconstruction

Mechanism. The BOS residual carries a massive activation the dictionary does not reconstruct: cosine 0.44 between input and reconstruction at position 0 against ≈ 0.93 elsewhere. Its norm dominates the pooled variance. **Evidence.** Including position 0 gives explained variance -3.5 ; excluding it gives 0.863 at mean L_0 85.4, matching the released specification. **Control.** Drop special-token positions; assert the expected explained variance and abort otherwise. Without that assertion every downstream number is computed on a broken hook point.

4.2 Perturbation magnitude

Mechanism. Ablation perturbs the residual stream by $a_f d_f$, and activation magnitude covaries with activation frequency, so raw effect sizes compare interventions of unequal size. **Evidence.** Median perturbation norm runs 3.60 in the rarest frequency decile to 14.76 in the most frequent. Uncontrolled, the rarest decile appears *worse* than a random dictionary; per unit norm it is better. The sign of the conclusion flips. **Control.** Report KL per unit perturbation norm throughout.

Corollary. Activation frequency predicts causal mass in neither the trained nor the control arm. The apparent frequency dependence in this setting is magnitude, not frequency.

4.3 Near-duplicate structure

Mechanism. Feature splitting produces decoder rows at high mutual cosine; ablating one member of such a pair leaves the concept partly represented. **Evidence.** Per-latent maximum cosine predicts causal mass in the released Gemma Scope dictionary ($\rho = +0.220$, $p = 0.0006$) and in neither of our trained arms ($+0.078$, $+0.060$), which have visibly less splitting (medians 0.207 and 0.122 against 0.272). **Control.** One-sided adjustment where the relationship is asymmetric. Symmetric residualisation is wrong here: fitting the control arm produces a slope of $+2.506$ against a predictor with $p = 0.82$, i.e. residualising against noise. Matching is infeasible against a random dictionary, which structurally cannot contain near-duplicates.

This artifact is conditional. Near-duplicate structure confounds causal-effect measurement *in dictionaries that have it*. That is more useful than a universal claim, because it comes with its own diagnostic.

4.4 Direct-path alignment: a sensitivity, not a confound

Mechanism. A trained decoder direction is aligned with directions the model writes to the residual stream, and those feed the unembedding. Next-token KL therefore rewards latents that short-circuit the network. We measure alignment as $\sqrt{d^\top C d}$ with C the embedding covariance — the spread of the direct logit contribution, since a uniform logit shift leaves the softmax unchanged.

Evidence. Alignment predicts causal mass at $\rho = +0.375$ in Gemma Scope and $+0.264$ in a dictionary we trained ourselves with a different architecture and roughly $40\times$ less data, with the control arm null in both cases ($-0.063, +0.004$). The single most causally powerful latent we found writes the letter k to the logits and carries $12\times$ the causal mass of the runner-up.

Why this is a sensitivity and not a confound. The trivially aligned latents are approximately the top 20 of 16,384, the 99.88th percentile. No sample a paper would realistically draw reaches them: the top twenty by alignment within a 240-latent sample sit near the 92nd dictionary percentile and are 90% word-like, under both stratified and uniform sampling. So at realistic sample sizes high-alignment latents are semantic, and adjusting alignment away removes signal rather than triviality. The correct statement is that ablation-based measurements are sensitive to unembedding alignment at $\rho \approx 0.3$, and that within realistic samples this alignment reflects semantic content.

An unexplained depth dependence. Because C is layer-independent by construction, the measure estimates the logit effect a direction *would* have if it reached the output unchanged, so it should predict best where fewest blocks intervene — deepest. It does the opposite. Alignment predicts causal mass with $\rho = 0.358$ at layer 5 and 0.339 at layer 12, against 0.031 at layer 19 and 0.145 at layer 24 ($n = 300$ per depth).

At $n = 300$ a Spearman correlation of 0.3 carries a 95% interval of about ± 0.10 , so the two shallow estimates are indistinguishable from each other and so, marginally, are the two deep ones. We therefore test the trend rather than describe the sequence: pooling by Fisher z gives $\rho = 0.349$ for the shallow pair and $\rho = 0.088$ for the deep pair, a contrast of $z = 4.74$ ($p = 2.1e - 06$), and a weighted regression of z on layer index has slope -0.0163 per layer where the measure’s construction predicts a positive slope. The pattern is a decrease with depth, not a mid-network dip with recovery.

The compressed-variance explanation is falsified: alignment spread is equal across depths (CV 0.1175 against 0.1091) and causal-mass spread is *greater* at layer 19. We report the dependence as unexplained; it is a violation of the measure’s own construction rather than an observation about the model.

4.5 Position in sequence

Mechanism. Effects are measured at each latent’s top-activating positions, and later positions carry more context and a sharper next-token distribution. **Evidence.** Pooled $\rho = -0.087$ ($p < 10^{-4}$) across 2,700 latents, reaching -0.207 in one dictionary and null in five. The sign is *negative*: earlier positions carry slightly more causal mass, opposite to the mechanism we proposed. On the endpoint sample the effect is not significant ($\rho = -0.084, p = 0.196$). **Control.** Record and report position; the effect is small enough not to threaten the endpoints here, but its mechanism is unexplained and its sign is not the obvious one.

5 Results

5.1 Endpoints

Table 1 gives the trained-versus-control comparison for three controls. The second and third rows share a dictionary, a feature list and an evaluation, so their ratio is a within-experiment

comparison	HL	95% CI	n
Gemma Scope vs untrained tied	2.269	[1.934, 2.638]	240
ours vs untrained tied	2.104	[1.826, 2.423]	240
ours vs soft-frozen	1.192	[1.026, 1.379]	240

Table 1: Causal effect per unit perturbation norm, Hodges–Lehmann ratios on uniform samples of live latents. The soft-frozen control has a trained encoder and a decoder constrained to cosine 0.8 of its random initialisation.

decomposition.

A trained dictionary beats an untrained one by $2.104\times$ and a soft-frozen baseline by $1.192\times$ ($p = 0.0213$). Most of the measured advantage is therefore attributable to the encoder; decoder freedom contributes the smaller factor. This holds despite the two arms differing sharply in decoder geometry — median cosine to initialisation 0.231 for the free decoder against 0.800 for the constrained one — and despite a reconstruction gap that grows monotonically over training ($+0.0064$ explained variance per million tokens, $p < 10^{-4}$).

5.2 The tail is not a population phenomenon

Across 9 Gemma Scope dictionaries spanning two depths, three widths and sparsity from L_0 22 to 445, the ratio of maximum to median causal mass within a fixed sample of 300 latents runs from $33\times$ to $464\times$, and the top 5% of latents carry between 36% and 73% of total causal mass. Every dictionary shows it.

But within any one dictionary the tail is a handful of latents. Removing the single largest takes the variance ratio from $42.1\times$ to $7.2\times$; the median ratio is unmoved. A bootstrap resamples those latents in and out while still treating them as population draws, so the jackknife curve is the honest instrument. At $n \approx 200$ the control arm’s 99th percentile has a bootstrap interval spanning $11.3\times$, and the derived “fraction above the control’s $p99$ ” runs [4.2%, 39.6%]: published aggregate comparisons on a few hundred latents cannot distinguish anything.

5.3 Effect decays with remaining depth, but not proportionally

Measuring (1) at t and $t+3$ across four depths at matched width and approximately matched sparsity gives $\text{KL}_{t+3}/\text{KL}_t$ of 0.384 (layer 5, 20 blocks remaining), 0.360 (layer 12, 13), 0.078 (layer 19, 6) and 0.021 (layer 24, 1), with $n = 300$ latents per depth. The effect itself is large and not in doubt: a span of $18\times$ from shallowest to deepest.

The *attribution* to propagation limits is much weaker, and we state it as such. We pre-registered three predictions of decreasing strength. A power law in remaining depth, fitted from layers 12 and 19 and tested out of sample with a factor-of-1.5 tolerance, predicts 0.760 at layer 5 and 0.003 at layer 24 against observed 0.384 and 0.021 — misses of $0.51\times$ and $8.4\times$, in opposite directions, so the curve is flatter than proportional at both ends. Two threshold predictions, >0.5 at layer 5 and <0.02 at layer 24, also both failed. Only bare monotone ordering survived, and on four points that is worth $p \approx 0.042$ under a random-permutation null.

So: $\text{KL}_{t+3}/\text{KL}_t$ declines with depth by a factor of 18; the propagation account predicts this ordinarily at $p \approx 0.042$ and is falsified as a proportionality in both directions. The decline is solid, the mechanism is not established.

Consequence for §4.4. Because this gradient is a propagation effect, it is not evidence that deeper layers are more direct-path dominated, and it must not be read as support for the alignment sensitivity.

5.4 Reconstruction improves without a matching causal gain

Evaluating the trained and soft-frozen arms at four training budgets gives HL ratios of 1.086, 1.098, 1.392 and 1.257 at 3, 6, 9 and 12 million tokens (slope +0.0270 per million, $p = 0.28$), while the explained-variance gap over the same range grows from 0.000 to 0.077 at $p < 10^{-4}$. The reconstruction gap grows significantly and the causal gap does not. With four budgets, 120 latents each, and non-monotone point estimates, this is consistent with a dissociation but does not establish one.

6 Limitations

One model. All measurements use `gemma-2-2b`. Whether the depth dependence in §4.4 is about relative depth or about this model is untested; Llama Scope dictionaries would answer it. **Training budget.** Our trained arms see 12M tokens against Gemma Scope’s billions, and the reconstruction gap is still widening at that point, so any claim about what decoder training does is scoped to this budget. **One corpus** for the trained arms (WikiText-103), and one evaluation corpus. **Ablation only.** These artifacts apply to metrics that zero or subtract a latent’s contribution and read a magnitude. They do not transfer directly to interchange-based scores such as RAVEL, which are match-rate accuracies; we make no claim about that result. Note that ablation, interchange and clamping all perturb along the same d_f , so “use interchange instead” is not a remedy — the axis that matters is magnitude versus rate. **Attribution not established.** The two SAE Bench metrics that fail reliability auditing are the only ablation-based metrics in that suite, but also the only set-ablation metrics; the properties are confounded and we do not claim their failure follows from these artifacts. **Unexplained.** The depth dependence of the alignment sensitivity, and the sign of the position effect.